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In [2]: # importing packages:
import cv2
import numpy as np
import os # helps to interact with the operating system
import matplotlib.pyplot as plt
import pickle # help to zip/compress the image file in byte(binary)
# format.
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In [3]: # accessing the images present inside "Emotion":
# getcwd:- fetching the current working directory
img_dir = os.path.join(os.getcwd(),"Emotion")
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In [4]: # to checking the directory where currently we are working:
print(os.getcwd())
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C:\Users\Admin\Downloads\Real Time Emotion detection

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In [5]: # creating a path to store the pickle files inside data folder.
img_final = os.path.join(os.getcwd(),'data')
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In [6]: # to store all the input(images) and output(labels) in different [].
images = []
labels = []
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In [8]: # performing image preprocessing and saving the files.
# listdir: to list all folders/file present in any directory.
for i in os.listdir(img_dir): # to iterate over "Emotion" folder
    for j in os.listdir(os.path.join(img_dir,i)): # to iterate over the sub folders ['angry','sad'] etc
        img_path = os.path.join(os.path.join(img_dir,i),j) # to fetch the image path
        img = cv2.imread(img_path) # to read the image
        img = cv2.cvtColor(img,cv2.COLOR_BGR2GRAY) # to convert 3 channel(BGR) to 1 channel (Grayscale)
        img = cv2.resize(img,(48,48)) # as all images have the shape of 48 X 48
        images.append(img) # to store processed images in the "images" list
        labels.append(i.split(' ')[0]) # to store the "file name (i.e: sad, angry etc)"
print(i,'---> Completed')
```

```
angry ---> Completed  
happy ---> Completed  
neutral ---> Completed  
sad ---> Completed  
surprised ---> Completed
```

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In [9]: set(labels)
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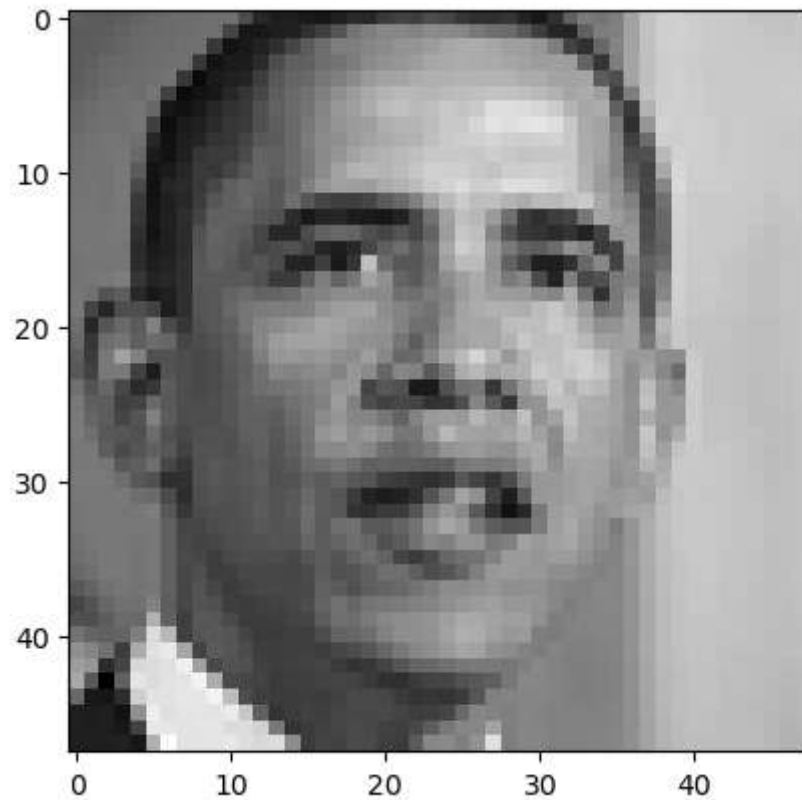
```
Out[9]: {'angry', 'happy', 'neutral', 'sad', 'surprised'}
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In [10]: images = np.array(images)  
labels = np.array(labels)
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In [11]: images.shape
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Out[11]: (16082, 48, 48)
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In [12]: # to plot an image after pre-processing.  
plt.imshow(images[1000], cmap='gray')  
plt.show()
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In [13]: # Pickling the images and labels files.  
# 'wb' - write binary as pickle file store only byte data  
# 'images.p' -> 'images' -> files name, '.p' -> pickle file's extension  
with open(os.path.join(img_final, 'images.p'), 'wb') as f:  
    pickle.dump(images, f)  
  
with open(os.path.join(img_final, 'labels.p'), 'wb') as f:  
    pickle.dump(labels, f)
```

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In [ ]:
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